O&M Incident Report & Postmortem: XG-440 Event (2025-11-21 to 2025-12-15)

Category: Procedures | Model: VFIP-OM-POSTMORTEM-2025 | Prepared: 2026-01-05

Description: Detailed operational incident report documenting anomaly detection on 2025-11-21 through remediation and partial recovery post-2025-12-08. Includes vendor firmware notes, change tickets, SCADA evidence, CMMS templates, runbooks, and settlement guidance for ERCOT.

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Executive Summary

This document provides a comprehensive operational incident report and postmortem for the XG-440 inverter firmware rollout that initiated anomalies on 2025-11-21 and continued through partial remediation actions through 2025-12-15. The incident caused an aggregate daily net energy drop across affected plants, triggered protective trips (notably TRIP-31 and TRIP-44), and required extensive corrective work orders. The incremental O&M cost attributable to the event totaled approximately $260,000. Root cause analysis links the outage to the vendor's firmware v3.15.0 rollout and an incorrect protective threshold change. This report contains vendor release notes, change ticket summaries, day-by-day chronology, SCADA-derived evidence tables, vendor communications, dispatch/runbook mappings, CMMS templates, and ERCOT settlement guidance for imbalance reconciliation.

Chronology of Events (Day-by-Day)

Note: Times are presented in local plant time (CST). Wherever possible, SCADA timestamps are quoted

in ISO 8601 format. "Affected fleet" refers to the set of XG-440 inverters within serial ranges listed in the change tickets section.

2025-11-21 — Detection & Initial Anomaly

05:12 — SCADA anomaly detection: sudden step-down in net MWh output across three plants (Plant A, B, C). Automated alarm group `ALM-GRP-INV_PERF_DROP` tripped. Initial symptom: simultaneous loss of ~28% of expected generation aggregated across affected inverters during the 05:00 - 06:00 interval.

06:05 — Field Ops dispatch: remote reboots attempted for 12 inverters; no restoration. Alarm log snapshots captured protective trip codes: TRIP-31 and TRIP-44 for multiple serials.

07:10 — Escalation to vendor support (ticket VF-20251121-001). Vendor acknowledges correlation with firmware rollout completed on 2025-11-20 and begins analysis. Temporary local curtailment disabled per dispatch instructions to prevent cascading trips.

Verification steps executed:

1. Confirm SCADA channel `INVERTER_AVAILABILITY` for each serial.
2. Extract net MWh by plant from `DAILY_ENERGY_SUM` table for the 24-hour window.
3. Snapshot logs: `INV_LOG_YYYYMMDD.log` from each affected inverter for 00:00–08:00.

2025-11-22 to 2025-12-08 — Escalation, Triage & Hotfix Preparation

2025-11-22 — Vendor provides preliminary root-cause hypothesis: a parameter change in v3.15.0 relating to anti-islanding/protection thresholds set lower than expected for certain grid conditions. Vendor identifies protective trip codes mapping to new threshold logic.

2025-11-24 — Wide-area non-synchronous grid events overlapped with a scheduled curtailment window (see Curtaiment Window Overlays section). The combination triggered TRIP-31 activation on 380 serial numbers across Plant A and B.

2025-11-28 — O&M raises corrective work orders and begins targeted manual rollback for 48 high-priority inverters where rollback is possible without field visit. Rollback procedure used: `FW_ROLLBACK_MANUAL_V3.15->3.14.2` following vendor rollback checklist.

2025-12-01 — Vendor posts hotfix candidate v3.15.1 (internal regression) and change ticket for planned staggered rollout 2025-12-08. Change ticket includes serial range exclusions and staging plan.

Verification & validation actions performed during this period:

1. Daily extraction of `DAILY_NET_MWH_DROP.csv` for trending.
2. Availability reconciliation with CMMS logs `WO_STATUS.history`.
3. Temporary dispatch rule implementation: disable auto-curtailment triggers when protective trip codes exceed threshold during active rollback windows.

2025-12-09 to 2025-12-15 — Partial Recovery, Hotfix Rollout, and Residual Actions

2025-12-08 — Vendor hotfix v3.15.1 staged rollout initiated; initial deployment to low-risk serial ranges. Hotfix included corrected protective threshold defaults and rollback safety checks for

grid detection logic.

2025-12-09 — Significant improvement noted: daily net MWh drop reduced by ~75% relative to peak-day loss. Inverter availability improved from an aggregate 71% affected to 92% available across the fleet within 48 hours post-hotfix for staged devices.

2025-12-12 — Full vendor hotfix rollout completed for all approved serial ranges. Field techs completed outstanding CMMS corrective work orders and parts requisitions on 2025-12-14.

Residual actions through 2025-12-15:

- Complete forensic log aggregation for 1,200 affected serials.
- Finalize cost accounting for incremental O&M.
- Issue final incident report and implement monitoring thresholds to detect recurrence.

SCADA-Derived Evidence

This section summarizes SCADA extractions, daily net MWh drop table, inverter availability by serial ranges, and annotated chart descriptions intended for inclusion in appendices for RAG ingestion. All data fields below are extracted directly from the SCADA historian schema.

Key SCADA Fields (Definitions)

Field Name	Type	Description
TIMESTAMP	datetime	ISO 8601 timestamp of sample (local).
PLANT_ID	string	Plant identifier (Plant A, Plant B, Plant C).
INVERTER_SERIAL	string	Unique inverter serial number.
ACTIVE_POWER_W	float	Instantaneous active power (W).
DAILY_ENERGY_MWH	float	Energy sum for the local day (MWh).
INVERTER_AVAILABILITY	percentage	Availability metric (0–100%) computed by uptimes in epoch.
PROTECTIVE_TRIP_CODE	string	Trip code logged by inverter firmware (e.g., TRIP-31).

Daily Net MWh Drop (Extract)

The table below shows extracted daily net MWh change computed as baseline expected minus observed generation. Baseline expected derived from rolling 30-day expected generation model field `MODEL_EXP_DAILY_MWH`.

Date	Plant A (MWh)	Plant B (MWh)	Plant C (MWh)	Aggregate Net Drop (MWh)
2025-11-20 (baseline)	2,100.6	1,725.4	1,012.1	—
2025-11-21	1,512.2	1,186.8	724.9	896.2

2025-11-22	1,431.9	1,159.0	712.3	1,014.9
2025-11-28 (peak)	1,250.3	998.5	654.7	1,334.6
2025-12-09 (post hotfix)	1,940.7	1,642.2	975.8	120.2
2025-12-15	2,020.1	1,692.3	995.6	28.1

Inverter Availability by Serial Ranges (Sample)

Availability computed over the 2025-11-21 → 2025-12-09 window unless otherwise noted.

Serial Range	Count	Availability Pre-Rollout (%)	Availability Post-Hotfix (%)	Primary Trip Codes Observed
XG-440-1000 → XG-440-1199	200	62.4	95.2	TRIP-31, TRIP-44
XG-440-1200 → XG-440-1399	200	68.0	94.8	TRIP-31
XG-440-1400 → XG-440-1599	200	75.3	97.1	TRIP-44, TRIP-18
Remaining fleet	600	89.9	99.2	TRIP-18

Annotated Chart Descriptions (for inclusion in evidence package)

Note: The following are text descriptions of annotated charts to be stored in the evidence archive. Use these descriptions when images are not available to RAG systems.

1. **Chart A — Aggregate Output vs Baseline (11/20–12/15):** Line chart showing baseline model expectation (dashed) and observed aggregate output (solid). Annotated spikes on 11/21 and plateau starting 11/24 vertical highlight indicates change ticket timestamp 2025-11-20 22:30 UTC. Annotation: "Initial anomaly detection window."
2. **Chart B — Per-Plant Daily Net MWh Drop Heatmap:** Heatmap with plants on Y-axis and date on X-axis; darkest cells at 11/28 indicating maximum drop. Annotation includes protective trip prevalence percentages per day.
3. **Chart C — Inverter Availability Timeline (Representative serial XG-440-1103):** Availability 0–100% stacked area with markers at firmware deployment and rollback times. Markers show TRIP-31 assignment at 05:41 11/21 and availability restoration after 12/09 hotfix.

Vendor Firmware Release Notes (v3.14.2 → v3.15.0)

Below are extracted and paraphrased vendor release notes and parameter changes relevant to this incident. Fields and parameter names map directly to vendor configuration keys.

v3.14.2 (Prior Stable)

Release Date: 2025-09-12

- Default protective thresholds: `ANTI_ISLAND_VOLTAGE_THRESHOLD=0.88_pu`, `FREQ_TRIP_LOW=57.5_Hz`.
- Known issues: Minor telemetry packet burst when concurrent comms loss; no protective trip changes.

v3.15.0 (Rollout 2025-11-20)

Release Date: 2025-11-20 — Full rollout initiated 2025-11-20 to selected serial ranges.

Key parameter changes introduced (excerpts):

Config Key	Old Value	New Value (v3.15.0)	Purpose
ANTI_ISLAND_VOLTAGE_THRESHOLD	0.88_pu	0.92_pu	Tighten island detection to reduce nuisance overproduction during weak grid.
FAST_RECONNECT_ENABLE	true	false	Prevent fast reconnection on transient faults.
TRIP_LOG_BUFFER_SIZE	50_entries	200_entries	Increased trip history retention.
GRID_RIDE_THROUGH_ALPHA	0.10	0.05	Changed smoothing parameter for ride-through detection.

Protective trip code mapping (new or updated behavior):

Trip Code	Meaning	Trigger Conditions (v3.15.0 logic)
TRIP-31	Voltage anti-islanding trip	Voltage below `ANTI_ISLAND_VOLTAGE_THRESHOLD` for >120 ms AND `GRID_RIDE_THROUGH_ALPHA` hysteresis window breached.
TRIP-44	Fast disconnect upon combined voltage/ frequency aberration	Voltage excursions `±0.10_pu` AND instantaneous frequency deviation >0.5 Hz within 200 ms.
TRIP-18	Internal comms watchdog timeout	Comms heartbeat absent for >3000 ms; increased sensitivity when `FAST_RECONNECT_ENABLE=false`.

Vendor recommended rollback conditions (excerpt): If >5% of farm inverters log TRIP-31 within a 4-hour window when no corresponding grid event is present, perform

emergency rollback to last stable r3.14.X and open incident escalation ticket (priority P1).

v3.15.1 (Hotfix, 2025-12-08)

Release Date: 2025-12-07 (staged rollout 2025-12-08)

- Reverted `ANTI_ISLAND_VOLTAGE_THRESHOLD` to 0.88_pu for serial ranges flagged as high risk. Introduced conditional param `GRID_THRESHOLD_PROFILE=dynamic` for region-specific behavior.
- Added safeguard: Do not auto-disable fast reconnect until operator confirmation when simultaneous trip counts exceed configured fleet percent `SAFEGUARD_TRIP_PERCENT=3`.
- Bugfix: resolved erroneous TRIP-44 logic related to frequency measurement aliasing under low-sampling telemetered environments.
- Rollback safety: firmware includes remote-disable of new threshold only if `OPERATOR_APPROVAL=true` (prevents blind mass rollbacks).

Relevant excerpts from vendor rollback guidance

```
# Vendor rollback checklist (extracted)
1. Query fleet with filter: FIRMWARE_VERSION == 'v3.15.0' AND SERIAL
IN configured_range
2. If TRIP-31 count > (fleet_count * 0.05) within 4 hours, mark for
immediate rollback.
3. Apply staged rollback in groups of <= 50 units to prevent power-
oscillation.
4. Monitor SCADA INVERTER_AVAILABILITY and DAILY_ENERGY_MWH after
each stage for 24 hours.
5. If rollback fails, escalate to engineering with full trip log
bundle: TRIP_BUNDLE_{timestamp}.zip
```

Vendor Change Ticket Summaries (Rollout & Hotfix)

This section contains summarized change ticket content extracted from the vendor portal for the relevant rollout and hotfix events.

Change Ticket: CT-20251120-R3.15

Field	Summary
Ticket ID	CT-20251120-R3.15
Scope	Firmware v3.15.0 staged rollout to XG-440 serials 1000–1999 (Plants A,B) and 3000–3199 (Plant C)
Planned Window	2025-11-20 00:00 — 2025-11-21 23:59

Change Owner	Vendor Release Engineering
Rollback Plan	Rollback to v3.14.2 if TRIP-31 rate >5% in any plant within 4 hours of deployment.
Notes	Exempt serials: 1500–1515 (sensitivity-tested earlier) — excluded from this rollout.

Change Ticket: CT-20251208-HOTFIX

Field	Summary
Ticket ID	CT-20251208-HOTFIX
Scope	Hotfix v3.15.1 targeted rollout for serials 1000–1999 and 3000–3199 (two-stage: low-risk then full)
Planned Window	2025-12-08 03:00 — 2025-12-08 18:00
Change Owner	Vendor Hotfix Team
Rollback Plan	Staged rollback to v3.14.2 only if trip count increases again post-hotfix; manual rollback require `OPERATOR_APPROVAL`.
Notes	Hotfix includes telemetry patch to increase trip logging fidelity (`TRIP_LOG_BUFFER_SIZE=200`), enabling better forensic analysis.

Vendor Communications Timeline

Consolidated timeline of key communications (tickets, teleconferences, and advisories).

Date	Sender	Recipient	Summary
2025-11-21 07:10	Vendor Support	O&M Ops	Acknowledgement of incident; correlates with 2025-11-20 rollout; P1 ticket opened VF-20251121-001.
2025-11-22 10:00	Vendor RE	O&M Ops	Preliminary analysis: anti-islanding thresholds tightened — possible cause. Requested trip bundle uploads.
2025-11-24 16:30	O&M Ops	Vendor Support	Provided SCADA extracts and set of failing serials; requested expedited rollback for critical serials.
2025-12-01 09:15	Vendor Release Engineering	O&M Leadership	Announced v3.15.1 hotfix candidate; scheduled 12/08 rollout window.
2025-12-08 03:05	Vendor Hotfix Team	O&M Dispatch	Hotfix rollout started; provided list of serials and stage IDs; monitoring ongoing.

2025-12-09 08:00	Vendor Account Manager	O&M VP	Status update: 75% reduction in net MWh drop observed in first 24 hours post-hotfix.

Escalation Records

Record of formal escalation actions taken.

- Escalation Level 2 opened 2025-11-23 at 14:20 (VF-ESC-20251123-L2): Requested engineering telepresence with vendor and inverter OEM to analyze protective trip logic and suggest temporary mitigations.
- Escalation Level 3 opened 2025-11-29 at 10:05 (VF-ESC-20251129-L3): Formal notice of potential contract breach due to energy losses; vendor provided hotfix timeline.

Dispatch Records & Runbooks

This section maps SCADA alarms to CMMS corrective work orders and details dispatch guidance used during the incident.

Dispatch Runbook: SCADA Alarm → Action

SCADA Alarm	Severity	Initial Field Action	CMMS WO Template
ALM-GRP-INV_PERF_DROP	High	Verify plant-level output; assign local tech to check inverter availability; attempt remote reboot (unless persistent TRIP).	WO-CORR-INV-001
ALM-TRIP-31-COUNT_HIGH	Critical	Quarantine affected serial range from auto-curtailment; perform staged rollback as per vendor rollback checklist; notify vendor.	WO-EMERG-ROLLBACK
ALM-TRIP-44-DETECT	Critical	Hold dispatch for frequency/voltage validation; if grid stable, prepare for manual firmware patch per vendor guidance.	WO-PATCH-INV
ALM-COMMS-LOSS	Medium	Investigate RTU/SCADA comms path; escalate to comms vendor if widespread.	WO-COMMS-RTU

Example Dispatch Log Entry (Representative)

```
Dispatch ID: D-20251121-0843
Plant: Plant A
Assigned Tech: FieldTech-07
Alarm: ALM-GRP-INV_PERF_DROP
Initial Action: Remote reboot attempted on serials
XG-440-1003,1008,1011 at 06:12. No change.
Follow-up: Dispatched FieldTech-07 for onsite check at 07:05.
Collected inverter logs, uploaded to bucket s3://ops-
```

```
evidence/2025-11-21/D-20251121-0843/
CMMS Reference: WO-CORR-INV-001 #WO-4567 created at 07:10
Notes: TRIP-31 present on 1003 log at 05:41. Vendor ticket
VF-20251121-001 opened.
```

CMMS Work Order Templates (Corrective Dispatch Entries & Parts Requisition)

Exact CMMS templates used for corrective dispatch and parts requisitions. These templates are encoded for direct copy-paste into the CMMS system (fields included).

WO Template: WO-CORR-INV-001 (Corrective Inspection & Remote Action)

```
WO_TYPE: Corrective
WO_TEMPLATE_ID: WO-CORR-INV-001
PRIORITY: P2
SUMMARY: Investigate inverter performance drop and collect logs
DESCRIPTION:
  - Inspect inverter(s) [INVERTER_SERIAL_LIST]
  - Attempt remote reboot via vendor API: vendor.reboot(serial)
  - Collect and upload inverter logs: /var/log/inverter/
INV_LOG_{serial}_{date}.log
  - Set field  in SCADA to 'UNDER_INVESTIGATION'
FIELDS:
  - PLANT_ID (required)
  - INVERTER_SERIAL_LIST (required)
  - ASSIGNED_TECH (required)
  - SCHEDULE_START (date/time)
  - EST_HOURS (float)
CHECKLIST:
  [ ] Remote reboot attempted
  [ ] Logs uploaded (path)
  [ ] Vendor ticket referenced
  [ ] Outcome updated in SCADA
CLOSE_ACTION:
  - Set INVERTER_AVAILABILITY updated
  - Link vendor ticket ID
```

WO Template: WO-EMERG-ROLLBACK (Emergency Firmware Rollback)

```
WO_TYPE: Corrective / Firmware Rollback
WO_TEMPLATE_ID: WO-EMERG-ROLLBACK
PRIORITY: P1
SUMMARY: Emergency rollback to last stable firmware for serial range
[SERIAL_RANGE]
```

```
DESCRIPTION:
  - Verify vendor approval for rollback (field: VENDOR_APPROVAL_ID)
  - Backup current state and export trip logs
  - Execute rollback procedure per vendor: execute_rollback(serial,
target_version)
  - Monitor INVERTER_AVAILABILITY for 24 hours post-rollback
FIELDS:
  - SERIAL_RANGE (required)
  - VENDOR_APPROVAL_ID (required)
  - TARGET_FW_VERSION (default v3.14.2)
  - STAGE_GROUP_ID (optional)
CHECKLIST:
  [ ] Rollback executed for each serial
  [ ] Post-rollback validation completed
  [ ] SCADA and CMMS records updated
CLOSE_ACTION:
  - Submit trip bundle to vendor
  - Finalize work order notes with energy delta
```

Parts Requisition Template: PR-INV-PARTS

```
PR_ID: PR-INV-PARTS
REQUESTOR: O&M Dispatch
PLANT_ID: [PLANT_ID]
PARTS:
  - PART_NUMBER: INV-CAP-BOARD-V2
    DESCRIPTION: Inverter capacitor board (stock replacement)
    QTY: [QTY]
  - PART_NUMBER: COMM-RTR-01
    DESCRIPTION: RTU communications router
    QTY: [QTY]
JUSTIFICATION: Parts requisition related to firmware-induced
protective trips causing increased field visits.
COST_CENTER: OPR-MAINT-2025
APPROVALS:
  - MAINT_SUPERVISOR
  - FINANCE_APPROVAL (if QTY*unit_cost > $5,000)
```

Cost Accounting for Incremental O&M ($260,000)

Detailed breakdown of the incremental operational costs incurred by the incident. All values are USD and rounded to the nearest dollar.

Category	Detail	Amount (USD)
Labor — Field Techs	120 tech-days @ $650/day (overtime included)	78,000

Labor — Dispatch & Engineering	800 staff-hours @ $85/hr	68,000
Vendor Support & Escalation Costs	Premium support hours, escalation fees	45,000
Parts & Logistics	Replacement boards, RTU routers, expedited shipping	28,500
Temporary Curtailment & Market Losses (estimated)	Unrealized energy value and imbalance exposure (see settlement guidance)	30,500
Total Incremental O&M		**260,000**

Notes on Allocation & Accounting

- Labor values include overtime multipliers where applicable (weekend emergency response factor 1.5x).
- Temporary curtailment losses are estimated based on day-ahead prices and observed net MWh drop; final settlement adjustments subject to ERCOT market processes (see next section).
- All vendor billed amounts are pending final invoice reconciliation and holdback provisions under SLA.

Settlement & Imbalance Charge Guidance (ERCOT)

Guidance for reconciling imbalance charges and settlement impacts in ERCOT following unplanned generation loss. This section includes calculation examples, required data fields for submission, and recommended reconciliation steps.

Key ERCOT Concepts

- Imbalance Energy (ACE): difference between scheduled deliveries (DAM/RTBM) and actual generation.
- Ancillary Services and Reg-Up/Reg-Down market settlements may apply if resource provided regulation or capacity obligations are missed.
- Settlement fields of interest: `SETTLEMENT_INTERVAL`, `METERED_MW`, `SCHEDULED_MW`, `IMBALANCE_MW`, `DAILY_EXCESS_CHARGE`.

Procedure for Reconciliation (Step-by-Step)

1. Extract metered data for the event period: field name `MTR_DATA_INTERVAL` from the meter system for every 5-minute interval.
2. Obtain day-ahead and real-time schedules: `DA_SCHEDULE_MW` and `RTBM_SCHEDULE_MW`.
3. Compute IMBALANCE_MW = SCHEDULED_MW - METERED_MW for each settlement interval.
4. Apply ERCOT imbalance price for interval: PRICE_IMB = RT_PRICE (from ERCOT market data). Interval charge = IMBALANCE_MW *

PRICE_IMB.

5. Aggregate charges and reconcile against bilateral contracts; prepare dispute package if metering calibration or SCADA data gaps exist.

Example Calculation (Representative)

```
Interval: 2025-11-21 05:00-05:05
DA_SCHEDULE_MW: 150.0
METERED_MW: 110.0
IMBALANCE_MW = 150.0 - 110.0 = 40.0
ERCOT_RT_PRICE = $25.50/MWh
Charge for interval = 40.0 MW * (5/60) hr * $25.50 = 3.4 MWh * $25.50
= $86.70
Note: Settlement uses MWh, convert minutes to hours as above (5/60
hours).
```

Data Fields to Reconcile for Disputes

Field	Source	Purpose
MTR_DATA_INTERVAL	Metering system	Definitive metered output for settlement.
SCADA_DAILY_ENERGY_MWH	SCADA historian	Cross-check against metered totals; identifies telemetry gaps.
SCHEDULE_MW_HISTORY	Scheduling system	Verify scheduled deliveries and any manual schedule changes.
TRIP_BUNDLE_{timestamp}.zip	Forensic logs	Correlate protective events to intervals with imbalance.

Recommendation: Submit a dispute package containing metered data, SCADA extracts, vendor trip bundles, and change ticket references to ERCOT settlements within the allowable contestation window (typically 20 business days for certain charges — refer to ERCOT protocol updates). Coordinate with Market Operations and Legal for recovery of imbalance penalties where vendor culpability is supported by logs.

Root Cause Analysis

This RCA uses the 5-Why method combined with log correlation, vendor parameter audit, and change control analysis.

Summary Finding

The primary root cause of the outage was an unverified parameter change introduced in vendor firmware v3.15.0 that increased the anti-islanding voltage threshold from 0.88 pu to 0.92 pu and reduced the grid ride-through smoothing parameter (GRID_RIDE_THROUGH_ALPHA). Under coincident low-voltage grid conditions and scheduled curtailment windows, the new thresholds caused a statistically significant increase in TRIP-31 activations, resulting in widespread

generation loss.

RCA Details (5-Why)

1. Why did generation drop occur? — Because XG-440 inverters logged TRIP-31 and disconnected.
2. Why did inverters log TRIP-31? — The anti-islanding voltage threshold was raised to 0.92 pu making them more sensitive to voltage depressions.
3. Why was the threshold raised? — v3.15.0 introduced parameter changes intended to harden anti-islanding detection without region-specific profiles.
4. Why was there no region-specific profile or gating? — Rollout plan did not include conditional profiles for ERCOT region or a pre-check for known low-voltage events overlaying scheduled curtailments.
5. Why were changes deployed without sufficient staging? — Change ticket CT-20251120-R3.15 excluded a small set of serials but did not mandate telemetry-based gating or mandatory rollback triggers activated early enough to stop the rollout.

Contributing Factors

- Insufficient region-specific configuration in firmware; one-size-fits-all parameter applied.
- Change control lacked early rollback automation and required manual approval for remote rollback (OPERATOR_APPROVAL barrier).
- Concurrent scheduled curtailment windows created network conditions that amplified threshold sensitivity.
- Telemetry sampling aliasing affecting frequency measurement accuracy contributed to TRIP-44 false positives in some serials.

Evidence Supporting RCA

1. Trip logs: >1,200 TRIP-31 entries from 2025-11-21 05:00–08:00 correlated to firmware version v3.15.0 in `TRIP_BUNDLE_20251121.zip`.
2. Change ticket CT-20251120-R3.15 documented parameter changes described in Vendor Release Notes section.
3. SCADA DAILY_NET_MWH_DROP showing anomaly onset exactly 12 hours after staged rollout start time stamps.

Recommendations & Next Steps

Actionable remediation and preventive controls to avoid recurrence.

Immediate Actions (Operational)

1. Enforce vendor rollback for all remaining v3.15.0 devices to v3.14.2 where hotfix cannot be applied within 24 hours (use WO-EMERG-ROLLBACK template).
2. Disable mass auto-curtailment triggers for affected plants until monitoring rules are updated.
3. Submit dispute package to ERCOT for imbalance charges where vendor logs demonstrate correlation with firmware-induced trips.

Short-Term Technical Controls (30–90 days)

1. Implement configuration gating: vendor to provide region-specific profiles and conditional rollout flags (GRID_THRESHOLD_PROFILE).
2. Require staged rollouts limited to <=50 inverters per stage with automatic rollback if TRIP-31 exceeds 3% in 2 hours.
3. Improve telemetry sampling rates for frequency measurement to mitigate aliasing (minimum 10 Hz sampling recommended).

Contractual & Vendor Escalation

1. Escalate to vendor executive account team for SLA enforcement and cost recovery discussions for incremental O&M costs ($260,000) and potential settlement of imbalance penalties.
2. Require vendor to provide extended warranty or credit for plant-level losses caused by defective firmware changes per existing contract clauses.

Monitoring & Governance

1. Update internal change control process: require vendor to supply pre-deployment simulation logs for any protective parameter changes with sensitivity analyses.
2. Establish permanent monitoring rule: if any protective trip code hits >2% fleet-wide in 1 hour, auto-quarantine rollout and alert leadership.

Appendix: Logs, Alarm Codes, and Evidence Extracts

Selected extracts, trip code mappings, and evidence file naming conventions for archiving and RAG ingestion.

Trip Code Quick Reference

Trip Code	Short Name	Primary Trigger	Recommended Action
TRIP-31	Anti-Islanding Voltage	Voltage < threshold for >120 ms	Check anti-islanding threshold; consider rollback/mechanical re-synchronization

TRIP-44	Voltage/ Frequency Combo	Combined voltage & freq excursion (fast)	Validate grid frequency source; inspect sampling fidelity
TRIP-18	Comms Watchdog	Heartbeat lost >3000 ms	Inspect RTU/comm path; verify TRU/ RTU firmware

Evidence File Naming Convention

Standardized naming for archival use and RAG indexing:

```
{PLANT_ID}_{INVERTER_SERIAL}_{YYYYMMDD}_{DATA_TYPE}.zip
Examples:
PlantA_XG-440-1003_20251121_TRIP_BUNDLE.zip
PlantB_SCADA_20251121_DAILY_NET_MWH.csv
```

Forensic Log Extract (Example)

```
[2025-11-21T05:41:23.142] SERIAL:XG-440-1003 FIRMWARE:v3.15.0
TRIP:TRIP-31 VOLTAGE:0.915_pu FREQ:60.02Hz GRID_RIDE_ALPHA:0.05
[2025-11-21T05:41:23.312] EVENT:DISCONNECT REASON:ANTI_ISLAND_TRIP
[2025-11-21T05:44:00.001] OPERATOR_NOTE:Remote reboot attempted;
result: no change; vendor ticket VF-20251121-001
```

Contact & Escalation List

Role	Name / ID	Contact	Notes
Vendor Account Manager	AM-VENDOR-01	am.vendor@example.com	Primary escalation
Vendor Release Eng	RE-VENDOR-02	re.vendor@example.com	Firmware patch owner
O&M VP	VP-OM	vp.om@example.com	Executive escalation

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Prepared by: O&M Incident Response Team | Model: VFIP-OM-POSTMORTEM-2025 | Version: 1.0

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